# Class 19 Learning Activity

### Matlab Work-Up

Before summarizing the assignment, the file, `act_19.m`, was created for performing the calculations. Within it a function of the same name was declared. Each of the blocks of code listed below are added to this file to complete the function for performing the calculations.

```
function act_19()
% Calculations for the Class 19 Learning Activity from REB, The Course
```

## Assignment Summary

Reaction

$$A + B \rightarrow 2Z \tag{1}$$

Rate Expression

$$r_1 = k_1 C_A C_B \tag{2}$$

Reactor: Non-adiabatic, liquid-phase SBSTR

Given Constants: $\Delta H_1$ = –184 kJ mol–1, $k_{0,1}$ = 4.866 x $10^{11}$ $m^3$ $mol^{-1}$ $min^{-1}$, $E_1$ = 74.1 kJ $mol^{-1}$, $V_0$ = 4.0 L, $C_{A,0}$ = 10 mol $L^{-1}$, $T_0$ = 20°C, $T_{ex,in}$ = 20 °C, $V_{ex}$ = 500 $cm^3$, $A$ = 550 $cm^2$, $U$ = 8480 J $m^{-2}$ $min^{-1}$ $K^{-1}$, $\dot{V}_{ex,in}$ = 1.0 L $min^{-1}$, $T_{ex,0}$ = 20 °C, $\tilde{C}_p = \tilde{C}_{p,ex}$ = 1 cal $g^{-1}$ $K^{-1}$, $\rho = rho_{ex}$ = 1 g $cm^{-3}$, $C_{B,in}$ 10 M, $T_{in}$ = 20 °C, and $T_{max}$ = 80 °C.

The given and known constants and the basis are converted to consistent units and the given equations are converted to anonymous functions and all are addded to `act_19`.

```
% global constants available to all functions
% given
dH_1 = -184000.0; % J/mol
k0_1 = 4.866E14; % L/mol/min
E_1 = 74100.0; % J/mol
P = 1.0; % atm
V0 = 4.0; % L
CA0 = 10.0; % mol /L
T0 = 20 + 273.15; % K
Tex0 = 20 + 273.15; % K
Vex = 500.0/1.0E3; % L
Aex = 550.0/1.0E4; % m^2
U = 8480.0; % J/m^2/min/K
Vdot_ex = 1.0; % L/min
Tex0 = 20 + 273.15; % K
Cp = 4.184; % J/g/K
rho = 1.0E3; % g /L
CBin = 10.0; % mol/L
Tex_in = 20 + 273.15; % K
Tin = 20 + 273.15; % K
Tmax = 80.0 + 273.15; % K
% known
R = 8.3136; % J/mol/K
Rpv = 8.206E-2; % L-atm/mol/K
% calculated
nA0 = CA0*V0;
nAf = 0.01*nA0;
```

Task: Analysis

- Item of Interest: $\dot{V}_{in,max} : \underline{T} \leq T_{max}$

## Preliminary Analysis

The concentration of A will decrease continually due to consumption by reaction. Taken alone, this would cause the rate to decrease. The temperature is expected to increase initially under the assumption that the heat released by the reaction is greater than the heat removed by the cooling water. Initially the increase in temperature is expected to predominate over the decrease in reactant concentration causing the reaction rate to rise.

While the reaction rate is rising, the rate of heat release will be increasing. The rate of heat removal by the cooling water will also increase, but by less than the heat generation. As time

progresses, the effect of decreasing concentration will become prodominant. That will reduce the rate of reaction and by extension, the rate of heat release. The temperature is expected to pass through a maximum and then decrease.

The optimum feed rate will keep the heat release low enough to allow the cooling water to remove sufficient heat that the maximum temperature reaches 80 °C.

(qualitative analysis)

## SBSTR Model Function

SBSTR Design Equations

$$\frac{dn_A}{dt} = -r_1 V \tag{3}$$

$$\frac{dn_B}{dt} = \dot{n}_{B,in} - r_1 V \tag{4}$$

$$\frac{dn_Z}{dt} = 2r_1 V \tag{5}$$

$$\frac{dT}{dt} = \frac{\dot{Q} - \dot{V}_{in}\tilde{C}_p\rho\left(T - T_{in}\right) - V r_1 \Delta H_1 + P\dot{V}_{in}}{V\tilde{C}_p\rho} \tag{6}$$

$$\frac{dT_{ex}}{dt} = -\frac{\dot{Q} + \dot{m}_{ex}\tilde{C}_{p,ex}\left(T_{ex} - T_{ex,in}\right)}{\rho_{ex}V_{ex}\tilde{C}_{p,ex}} \tag{7}$$

$$\frac{dV}{dt} = \dot{V}_{in} \tag{8}$$

SBSTR Model Variables: $\underline{t}$, $\underline{n}_A$, $\underline{n}_B$, $\underline{n}_Z$, $\underline{T}$, $\underline{T}_{ex}$, and $\underline{V}$.

Additional Computable Unknowns: $r_1$, $k_1$, $C_A$, $C_B$, $\dot{n}_{B,in}$, $\dot{Q}$, and $\dot{m}_{ex}$.

$$k_1 = k_{0,1}\exp\left(\frac{-E_1}{RT}\right) \tag{9}$$

$$C_i = \frac{n_i}{V} \qquad i = A \text{ and } B \tag{10}$$

$$\dot{n}_{B,in} = C_{B,in}\dot{V}_{in} \tag{11}$$

$$\dot{Q} = UA_{ex}\left(T_{ex} - T\right) \tag{12}$$

$$\dot{m}_{ex} = \rho_{ex}\dot{V}_{ex,in} \tag{13}$$

Uncomputable Unknowns (Received as Arguments):

- Coupled Unknown: $\dot{V}_{in}$

IVODE Solver Requirements

1. Initial values

$$t_{iv} = 0 \tag{14}$$

$$n_{A,iv} = n_{A,0} = V_0 C_{A,0} \tag{15}$$

$$n_{B,iv} = n_{Z,iv} = 0 \tag{16}$$

$$T_{iv} = T_0 \tag{17}$$

$$T_{ex,iv} = T_{ex,0} \tag{18}$$

$$V_{iv} = V_0 \tag{19}$$

2. Stopping criterion

$$n_{A,f} = 0.01 n_{A,0} \tag{20}$$

3. SBSTR derivatives function that

    a. receives values of the SBSTR model variables ($t$, $n_A$, $n_B$, $n_Z$, $T$, $T_{ex}$, and $V$) at the start of an integration step

    b. can access the current value of $\dot{V}_{in}$

c. calculates the additional unknowns using equations (2) and (9) through (13)

d. evaluates and returns the SBSTR design equation derivatives using equations (3) through (8)

To facilitate making $\dot{V}_{in}$ available to the SBSTR derivatives function, the global variable, `g_Vdot` is declared and set to `nan` initially. Then the SBSTR model and (residuals/derivatives) functions are added to `act_19`.

```
% global variable for the current feed rate
g_Vdot = nan;

% SBSTR reactor function
function [t, nA, nB, nZ, T, Tex, V] = sbstr_model_variables(Vdot)
    % make Vdot available to the derivatives function
    g_Vdot = Vdot;

    % define the initial values
    ind_0 = 0;
    dep_0 = [nA0, 0, 0, T0, Tex0, V0];

    % define the stopping criterion
    stopVar = 1;
    stopVal = nAf;

    % solve the SBSTR design equations
    odes_are_stiff = true;
    [t, dep, success, message] = solve_ivodes(ind_0, dep_0, stopVar...
        , stopVal, @sbstr_derivatives, odes_are_stiff);

    % check for solver issues
    if ~success
        disp('')
        disp(["SBSTR model function issue: ", message])
        disp('Press any key to continue or Ctrl-C to exit.')
        pause
    end

    % return the sbstr model variables as separate arrays
    nA = dep(:,1);
    nB = dep(:,2);
    nZ = dep(:,3);
    T = dep(:,4);
```

```
        Tex = dep(:,5);
        V = dep(:,6);
    end

    % SBSTR derivatives function
    function ddt = sbstr_derivatives(~, dep)
        % extract the dependent variables
        nA = dep(1);
        nB = dep(2);
        T = dep(4);
        Tex = dep(5);
        V = dep(6);

        % calculate the additional unknowns
        nBin = g_Vdot*CBin;
        k1 = k0_1*exp(-E_1/R/T);
        CA = nA/V;
        CB = nB/V;
        r1 = k1*CA*CB;
        Qdot = U*Aex*(Tex - T);
        mDot_ex = Vdot_ex*rho;

        % evaluate the derivatives
        dnAdt = -r1*V;
        dnBdt = nBin - r1*V;
        dnZdt = 2*r1*V;
        dTdt = (Qdot - g_Vdot*Cp*rho*(T - Tin) - r1*V*dH_1...
             + P*g_Vdot*R/Rpv)/(V*Cp*rho);
        dTexdt = -(Qdot + mDot_ex*Cp*(Tex - Tex_in))/(rho*Vex*Cp);
        dVdt = g_Vdot;

        % return the derivatives as a column vector
        ddt = [dnAdt; dnBdt; dnZdt; dTdt; dTexdt; dVdt];
    end
```

## Coupled Unknowns Residuals Function

1. receives a guess for the coupled unknown $\dot{V}_{in}$ as an argument
2. calls the SBSTR model function using the guess as the argument to get $\underline{t}$, $\underline{n}_A$, $\underline{n}_B$, $\underline{n}_Z$, $\underline{T}$, $\underline{T}_{ex}$, and $\underline{V}$
3. evaluates and returns coupled unknowns residual

$$\dot{V}_{in} \; : \; \max\left(\underline{T}\right) = T_{max} \quad \Rightarrow \quad 0 = \max\left(\underline{T}\right) - T_{max} = \epsilon \tag{20}$$

The coupled unknowns residuals function is added to `act_19` next.

```
% coupled unknown residual function
function epsilon = coupled_unknown_residual(guess)
    % solve the sbstr design equations using the guess
    [~, ~, ~, ~, T, ~, ~] = sbstr_model_variables(guess);

    % evaluate and return the residual
    epsilon = max(T) - Tmax;
end
```

## Items of Interest Function

1. Define an initial guess for $\dot{V}_{in}$
2. Call an ATE solver passing the guess and the coupled unknown residual function as arguments to get the true value of $\dot{V}_{in}$ and report the result.
3. Optionally calculate the maximum temperature to ensure it was 80 °C.

To complete `act_19`, the items of interest function was added along with a line of code to call it and perform the calculations.

```
% items of interest function
function items_of_interest()
    % guess the optimum flow rate
    % Vdot_guess = 1.0; % failed to converge
    Vdot_guess = 0.01;

    % solve the implicit equation for the optimum flow rate
    [soln, success, message] = solve_ates(@coupled_unknown_residual...
        , Vdot_guess);
    Vdot_opt = soln;

    % check for solver issues
    if ~success
        disp('')
        disp(["Issue solving the implicit ATE: ", message])
        disp('Press any key to continue or Ctrl-C to exit.')
        pause
```

```
        end

        % solve the sbstr design equations using the optimum Vdot
        [t, nA, nB, ~, T, Tex, V] = sbstr_model_variables(Vdot_opt);

        % find the maximum T and concentration of B
        maxT = max(T) - 273.15;
        max_CB = max(nB./V);

        % tabulate, show, and save the results
        item = ["Optimum Feed Rate"; "Maximum Temperature"...
            ; "Maximum Concentration of B"];
        value = [Vdot_opt; maxT; max_CB];
        units = ["L/min"; "°C"; "M"];
        resultsTable = table(item, value, units);
        disp('')
        disp(resultsTable)
        disp('')
        writetable(resultsTable, 'act_19_results.csv')

        % for discussion, plot fA and T vs t
        figure
        plot(t,100*(nA0 - nA)/nA0, 'LineWidth', 2)
        xlabel('Time (min)','FontSize', 14)
        ylabel('Conversion of A (%)','FontSize', 14)
        set(gca,'FontSize', 14)
        saveas(gcf, 'act_19_fA_vs_t.png')

        figure
        hold on
        plot(t, T-273.15, 'LineWidth', 2)
        plot(t,Tex-273.15, 'LineWidth', 2)
        hold off
        xlabel('Time (min)', 'FontSize', 14)
        ylabel('Temperature (°C)')
        legend({'Reacting Fluid', 'Cooling Water'}, Location='best')
        saveas(gcf, 'act_19_T_vs_t.png')
    end

    % perform the calculations
    items_of_interest()
end
```

## Results

The calculations were performed by calling the function, `act_19`. The solver did not converge using an initial guess of 1.0 L min$^{-1}$ for $\dot{V}_{in}$, but changing the guess to 0.01 L min$^{-1}$ resulted in convergence.

```
>> act_19
                item                    value        units
    ----------------------------    ----------    -------

    "Optimum Feed Rate"               0.015927    "L/min"
    "Maximum Temperature"                   80    "°C"
    "Maximum Concentration of B"    0.00013053    "M"
```

## Discussion

To check the maximum conversion, he SBSTR design equations were solved using the optimum feed rate. The resulting temperature and conversion profiles are shown in Figures 1 and 2, respectively.

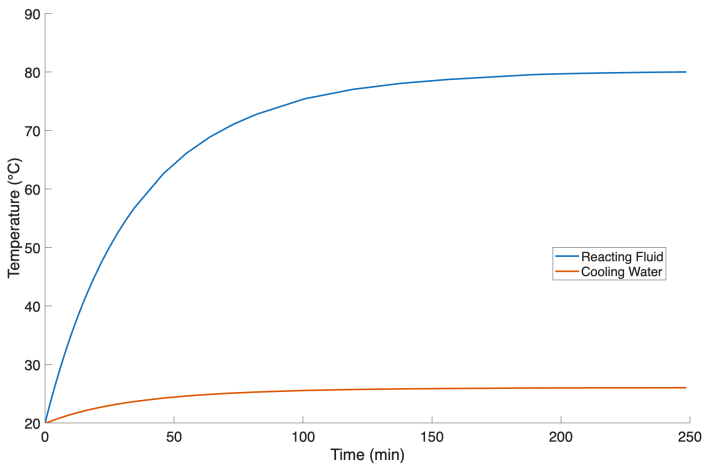



Figure 1: Temperature profiles during the addition of reagent B to the reactor.

As noted in the preliminary analysis, the temperature does rise while the concentration of A decreases. The temperature profile in Figure 1 confirms that the reacting fluid temperature never exceeds 80 °C. Figure 2 shows that the conversion increases linearly. This indicates that the reaction is likely extremely fast so that the B is converted almost instantly as it is added.

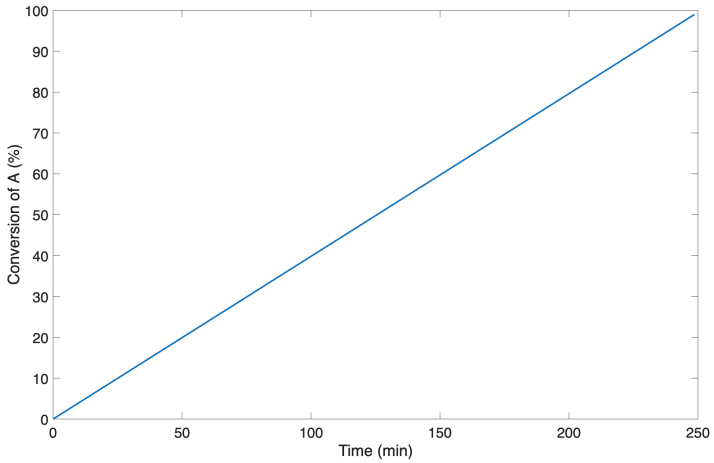



Figure 2: Conversion of reagent A during the addition of reagent B to the reactor.

Since it is added at a constant rate, the conversion increases linearly. This is confirmed In that the maximum concentration of B over the whole process is less than 0.001 M.

The temperature is well-controlled during the process. This suggests that if additional cooling capacity was added to the reactor, the feed rate could be increased which would shorten the processing time.

## Summary

To limit the maximum temperature to 80 °C, a feed rate of 0.016 L $min^{-1}$ must be used.